

ROAD HEALTH & CITIZEN GRIEVANCE REPORT

CONDITION: FAIR

REF: RHA-20260903-787F87

Automated computer-vision assessment of road surface condition, defect extent and remediation urgency, prepared for municipal grievance submission.

ROAD HEALTH INDEX

85.46
OUT OF 100

FAIR

DEFECTS DETECTED

5

3 potholes, 2 alligator cracks

DEFECT AREA RATIO

14.54%

of the photographed surface

HIGHEST SEVERITY

High

Pothole - tag #1

ROAD RISK ASSESSMENT

ROAD RISK INDEX

32.4

MODERATE RISK

Priority-3

Remediate within 30 days

REMEDIATION URGENCY

32.4 / 100

LOW

MODERATE

HIGH

CRITICAL

Extent

58/100

14.54% of the frame damaged · weight 50%

Worst defect

61/100

6.08% of the frame in one box · weight 30%

Density

62/100

5 defects detected · weight 20%

Weighted mean of the three components above, scaled by a hazard factor of 0.89 (how dangerous the detected classes are, area-weighted) and a confidence factor of 0.61 (mean detection confidence, floored at 0.50).

GEOSPATIAL & CAPTURE TELEMETRY

Date & time	03 Sep 2026, 03:16 PM
Coordinates	17.494800 N, 78.399600 E
Region	Hyderabad, Telangana, India
Source image	road_photo.jpg (1600 x 1000 px)
Evidence SHA-256	b9483e7a8780ec49c6183268d61cfc6c...
Map link	Open this location in Google Maps

MUNICIPAL JURISDICTION & ROUTING

Recipient body	Greater Hyderabad Municipal Corporation (GHMC)
Engineering wing	Roads & Maintenance Division
Grievance tier	Priority-3 - Remediate within 30 days
Dominant defect	Pothole
Priority action	Hot-mix / cold-mix asphalt fill with mechanical compaction
Submission status	Prepared by the citizen - not yet filed with the authority

Official Citizen Grievance Statement

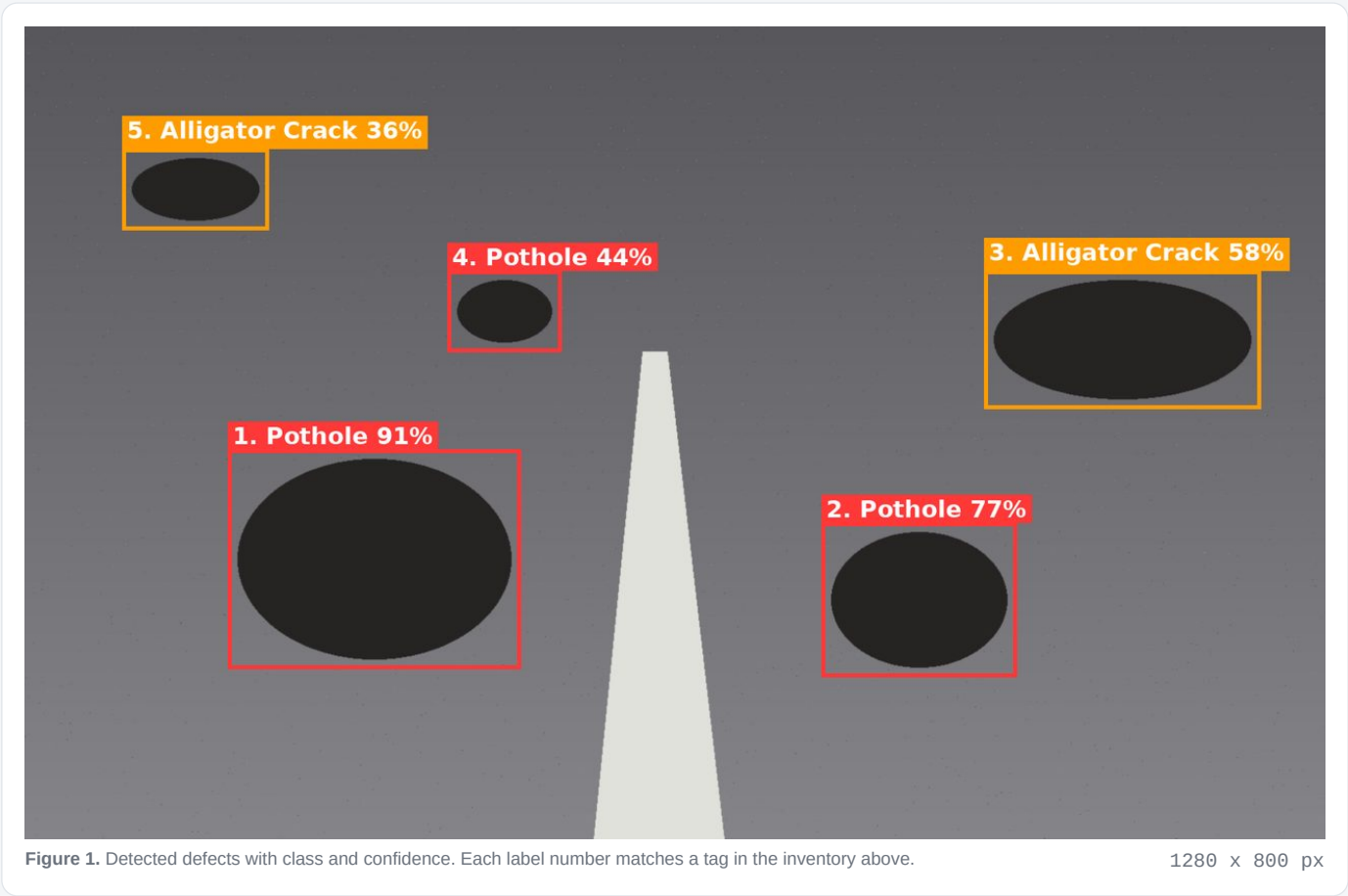
Road surface defects were detected in a photograph captured at latitude 17.494800, longitude 78.399600. Automated analysis identified 3 potholes and 2 alligator cracks, covering approximately 14.54% of the photographed road surface. The computed Road Health Score is 85.46 out of 100, indicating a FAIR road condition. The estimated Road Risk Index is 32.4 out of 100, placing this location in the MODERATE risk band; the suggested grievance tier is Priority-3 (remediate within 30 days). The most serious individual defect is rated high severity. The Greater Hyderabad Municipal Corporation (GHMC) is kindly requested to inspect the location and undertake the necessary repair work.

Addressed to: Greater Hyderabad Municipal Corporation (GHMC)

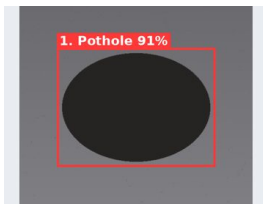
DEFECT INVENTORY & RECOMMENDED REMEDIATION

TAG	CLASSIFICATION	DETECTION CONFIDENCE	SEVERITY	RECOMMENDED REMEDIATION
#1	Pothole	91%	High	Hot-mix / cold-mix asphalt fill with mechanical compaction
#2	Pothole	77%	Medium	Edge trimming, tack coat and patch sealing
#3	Alligator Crack	58%	High	Bituminous slurry seal over the cracked area
#4	Pothole	44%	Minor	Localised surface levelling during the next maintenance round
#5	Alligator Crack	36%	Minor	Crack sealing during routine maintenance

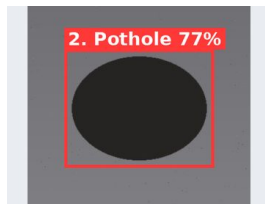
VISUAL EVIDENCE



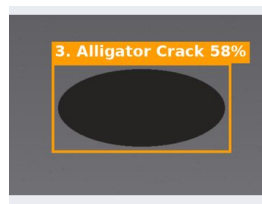
ISOLATED DEFECT CLOSE-UPS



Tag #1 - Pothole
High - 91% confidence



Tag #2 - Pothole
Medium - 77% confidence



Tag #3 - Alligator Crack
High - 58% confidence



Tag #4 - Pothole
Minor - 44% confidence



Tag #5 - Alligator Crack
Minor - 36% confidence

METHODOLOGY & MODEL TELEMETRY

Detection. trained object-detection model best.pt, single-pass CPU inference at 640 px, confidence threshold 0.25, NMS IoU 0.45. Classes: pothole, alligator_crack.

Health score. Defect area was measured by the 'sum' method - summed bounding-box area 14.54%, overlap-corrected union area 14.54% - as a share of the 1600 x 1000 px frame. Road Health Score = 100 - defect percentage = 85.46.

Risk index. $(0.50 \times \text{extent} + 0.30 \times \text{worst defect} + 0.20 \times \text{density}) \times \text{hazard } 0.89 \times \text{confidence } 0.61 = 32.43$, placing this location in the Moderate band (Priority-3, Remediate within 30 days). Severity of a single defect is banded by the share of the frame its box covers.

Limitations. Every threshold, weight and band used above - the Good/Fair/Poor/Dangerous score bands, the severity bands, the class hazard weights and the risk bands - is defined by this project team for the purpose of this tool. None of them is taken from IRC, MoRTH, GHMC or any other official standard. A photograph carries no depth or scale, so 'severity' reflects how much of the frame a defect occupies, not how deep it is, and the same defect photographed from a different distance will score differently. The suggested remediation is a starting point for the engineering wing, not an engineering directive. This document is an automated visual estimate from a single photograph and does not replace physical inspection by a qualified engineer.